

Episode #5 —Karl Friston, Neuroscientist, University College London

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[Music] i would like to uh introduce you to the viewers as a professor of neuroscience at the university college of london and you've made substantial theoretical contributions to neurobiology including the free energy principle and you're a fellow of the academy of medical science past president of the international organization of human brain mapping fellow of the royal society of biology weldon memorial prize and medal for contributions to mathematical biology honorary doctorates including one from the university of zurich i have to say that's quite impressive cv that you have there so i think i'd like to start with a few preliminary questions and it really is how in your view as a preeminent neuroscientist view the early and late childhood reading as creating models or frameworks of images concepts of ideas is that something that you have worked into your theory um in a in a general way absolutely um you know if you cast our job as neonates as infants as building models of the lived world and our lived world comprises other creatures like us and in particular when we're very young mum and dad and siblings and the like and then it becomes imperative to understand a world that is in cultured and by that i mean things that come along constructs that shape that world um and make it sensible in the sense of making sense of the world and language is is probably what you know that's a post-child of that um thing that has to be learned has to be instilled in your model of the world and in particular making sense of diabetetic interactions right from birth you know first of all just sorting out whether um your mom is me or i am somehow different from mother am i in control of me am i in control of mother developing a sense of self so you know the the act of inferring that mother is somehow separate from me another creature like me but not me upon all sorts of communication or initially affinity touch but within months we now have the machinery of language um and that machinery really does fast track that model learning that um you know sort of building apt constructs and models that you can predict what's going to happen next and indeed of course predict what you're going to

do next what you're saying next right your intentions your propositional stances and the like so language then um you know you just look at a school curriculum from you know from the start plays a really central role um you know in in scaffolding that model learning that structure learning that model building you know language uh is is a fascinating terrain and of course it goes from writing to printing to the current uh internet way of which language and image is combined on screens and so i think there's been quite a different approach and understanding of language and accessibility to language uh and has that had an impact in terms of just the way our brains now generate models of reality is that substantially different than before the invention of the printing press go back to the ancient greeks where it was all oral storytelling that's a a really interesting question um so i'm just thinking in my mind so from from my point of view um your language is the vehicle that provides a structure to the narratives that we bring to the table to understand what's happening you're on the outside of our brains beyond our skulls and what you're effectively asking is is the written word is that point of reference that consolidation that crystallization of the the cues and the structure um instrumental in disseminating and improving the learnability of these narratives i think so absolutely yes yes i hadn't really thought about that would be fascinating to model that um you know you know because language prior to the written word yeah lives in my head and your head but once it gets your poppers third um there's a there's a reference and a continuity there that i would imagine would be best seen as something that is sort of self-selecting in terms of hierarchical covolution that enhances the learnability of language of language in and of itself enhances the learnability of your generative models and your inference and you're making sense you know making sense of the world so yeah so so in a sense to talk about communication as a heart of the language really have to i guess look at the medium in which the communication has occurred because the to use marshall mcluhan the media may be the message uh that the way we construct our models will depend very much on the boundaries of that media the amplification of that medium and and the way that uh we can use it for our purposes and purposes of the community yeah yeah uh i mean that speaks this notion or a notion of a shared narrative you know if i if i'm using my model of the world to understand you and make sense of you effectively if i assume that you've got the same kind of model that i'm using then our narrative is is literally a shared narrative it's it's it's you know it's not mine nor yours it is something that we're sort of um committed to together so that's your that shared narrative um is something that we have to aspire to and learn together so your it is defined cooperatively again in an uncultured in an uncultured way and the modalities and the shared

commitments to the way that we use whether it's in print or indeed you know you one could even argue that mathematics is uh you know a kind of language that only works because there's a shared commitment to using this modality these shapes these kinds of sensory cues to enable me to infer what you are thinking and what you're going to say next the uh the other preliminary thing i wanted to raise is i've been fascinated because i've seen a number of your interviews particularly about the markov blanket uh which is probably would take a whole show just to go through that and i'll let you introduce what that concept is but whether the kind of generating models that we get as a child from our reading experience could it be said that those form a kind of markov blanket in a sense explains to me cognitive biases a cognitive bias are the defenses against basically contrary or hostile images or thoughts or ideals or principles and so we have a cognitive bias we have for example confirmation bias we look for those articles those books those peoples that confirm our beliefs and values yeah no i think that's that's an excellent point yep uh i mean you know cognitive biases you could actually read as as sort of um base optimal biases that they you know so if you subscribe to the free energy principle that rests upon markov blankets you are in the game all of us anything that exists is in the game of securing evidence for our own existence which just means evidence for our models of the world so you're going to seek out confirmation and if you're successful in seeking out confirmation evidence that you exist and you've got a good model in the world you exist for longer so it's almost tautological truism that what you've just said is is absolutely right one of the things that you were i heard you i say in another interview and i love this in the context particularly of fiction and novels if you are constantly surprised you've had a bad model of the world so in a way you you can judge the cognitive equipment that a person has how it's been shaped by how surprised they are so i i guess the wider your reading is as a child in many different genres and styles and periods the less surprised you're going to be when you see something because the range of activity that would have to surprise you would be vastly greater yes no absolutely in fact um we actually use um measures of surprise as um measured by uh electrical brain responses from people with autism or schizophrenia um uh you know just to do just to get a physiological handle on this so you're you know people do actually use surprise or oddball responses to sort of phenotype um various conditions uh where people may have difficulty inferring what's going on going going on around them so uh but you you also um bring to the table the breadth of reading the um the different kinds of outcomes and exchanges with your environment cultured or not and will certainly shape and be essential um in determining the shape of the model that you're you know that you're um you take forward

into later life and if you have a um a model that generalizes to different situations that you can predict synthesize in a way that renders it unsurprising when sufficient information has been secured then you're certainly going to have a better model that is more context sensitive and will underwrite a longevity in a changing world or a very context-sensitive world i'm just wondering where uh again getting back to uh reading books and fiction in particular where imagination and creativity fit into your view of modeling uh and generating models because although you may be surprised surprise for example if you're an improv musician is actually a good thing yeah you can riff off of that so that you you in a sense you invite the surprise and when you're writing fiction uh in some ways you're looking for the because that's what the reader wants the reader is looking to be surprised because if it's just all predictable they say the book is boring yep and another another excellent point and a key issue in the philosophy of the free energy principle which rests upon something called the dark room problem you know if we just want to minimize surprise why don't we go and seclude ourselves in a dark room and stay there forever and then we'll never be surprised um but that sort of kind of misses the point um if we are curious creatures that are always in the compelled to resolve uncertainty then the resolution of surprise or expected surprise as a consequence of our actions um is going to be one of the primary drives uh for surprise minimization so we have to look for the opportunity to minimize surprise so we listen to stories we watch um tv series we read books because we know at the end there will be a denouement there will be a resolution of uncertainty and if you read uncertainty's expected you know there's going to be a wonderful opportunity for that aha moment when you've your surprise mathematical kind of surprise is actually is actually resolved and i think it's a great explanation for music you know that you set up an uncertainty sure in terms of the way that this rift could go or you know what's going to happen in the next few bars and then you're compelled to continue listening actively listening in order to resolve that uncertainty but it's a resolution of uncertainty that drives novelty-seeking and curious behavior and i repeat if uncertainty is expected surprise a resolution of uncertainty is a minimization of this mathematical kind of surprise right so one one of your papers i would certainly recommend uh that viewers have a look at is in neuroscience and biobehavioral reviews november 2020 issue it's a brilliant paper that looks very closely at the whole notion of generating models and the commitment of the brain as this constructive organ that we are constructing models and testing them against reality at every stage and when somehow the model isn't quite working well as i understand from the paper this then we go into a question and answer what are we missing in the context of this thing why is it a little bit off or is our

model off do we have to adapt it in order that we see more clearly the reality that's there in other words it's not anything other than we have low information and what we're seeking is to increase the level of information so that we have a better understanding and explanation of what happens next or what may happen next yeah that is a beautiful summary of the essence of active inference absolutely and i think it touches on you the last exchange in relation to you know why we are compelled to be curious why are we curious creatures and and that it all boils down to this minimization of um a resolution of uncertainty um entailed by our current models and states of affairs in the world so how would you resolve that you resolve that uncertainty by gaining information how do you do that in the optimal way well you you basically um respond to epistemic affordances you you know where to find um the right kind of information it's going to resolve your uncertainty so you listen to that news channel you read that author um you you you you go to that wikipedia service so you you you you know what you don't and asking questions querying the world is the optimal way to minimize your uncertainty so you can actually write this down in terms of a you know an expected free energy or an expected surprise and if again um i ask you to read expected surprise of as uncertainty minimizing this expected free energy following an action following a move on is um inevitably compels you to gain the most information that resolves your uncertainty about the world and interestingly of course the the degree of epistemic attractiveness or affordance of a particular move a particular question a particular query the choice of things that you might read and depends upon your current belief state it depends upon what you know what you don't know so of course you could get bored when you've been foraging epistemically and there's no more uncertainty so then you'd move on to move on to something else and you know our work suggests that that sort of epistemic motivation that in um neuro robotics for example is known as intrinsic motivation just you know what would happen if i did that resolving uncertainty um about the way that the world works that's probably much much more important than simple sort of utilitarian pragmatic parents like your being rich or um if you're a monkey getting fruit fruit drops in in an experiment so i think that resolution of uncertainty that information gain um is underwrites most sentient behavior in systems like like us um and you know not every system needs this you know viruses and thermostats wouldn't have this epistemic but i think at a certain level when your generative model is sufficiently deep and prospective and it has to be in order to cope with living in a world that is constituted by other things like you that are also very deep in prospective you know then you become equipped with a model that entertains the consequences of your action and of course if you measure the

consequences in terms of what matters then the only thing that matters is getting down that expected surprise that expected free energy which just is being curious yeah i i i'm glad you you mentioned the word curious because it seems to me that curiosity is not equally distributed some people seem to be far more curious than others maybe to their darwinian disadvantage but there there is a range of curiosity and i'm wondering if that correlates with creativity and imagination yeah i'm sure yeah i'm sure it does um you know so technically um what i think we are talking about here is in the cognitive neurosciences something called structure learning so we're talking about the structure of our models um if you're in machine learning and you're doing deep learning for example how many how many layers would you have in a hierarchical model if you're a neurodevelopmental psychologist and you're you want to know what particular at what point different levels in the auditory hierarchy or the language system come online and what we're talking about is building models literally by wiring in different components so but you have to know the structure and of course you can't know the structure all you can do is explore different structures see whether they're fit for purpose see whether they explain the data at hand but and therefore you have to explore different models and i think this is where creativity and imagination comes in it's the ability to um think about counterfactual structures in the world you know what would happen if the world was like this or we or like that was counterfactuals included these differential hypotheses you can then use your sensory exchanges with the world or with others or with books to actually evaluate one um model in relation to another literally using the the the model evidence the evidence that model inherited in in the sensory data so that's how statisticians and people in machine learning actually optimize the structure of their models and i think that's exactly what we do during neurodevelopment and and subsequent experience dependent learning and what would that look like it would look like um you know a creature a system an agent who entertains in a very creative way different compositions different structures different constructs um that could possibly explain their world and then testing out the evidence for these different for these different uh constructs and if you're an artist the way that you might do that is by looking at the reactions of other people or indeed your own reactions to something that's now out there it's being realized physically in terms of a story or in terms of in terms of a picture so imagination would be if you like um sort of online uh structure learning and creativity um that allows you just um on the fly moment to moment just consider different hypotheses about what's going on now but exactly the same sort of mathematical imperative i think underlies um the building of the structures themselves during

neurodevelopment and during our ongoing um reorganization of of our brains and the concepts are entailed by by that structure learning yeah counterfactuals are interesting and i'm wondering if that's something that is taught along with early childhood reading as well is that in a way you could say shakespeare may have had a right is that you know all life is a stage and we're all actors on it and that childhood reading is a series of scripts in which you see the human drama unfolding in front of you and by understanding that you understand what's expected of you socially you understand what are the boundaries of behavior what you can expect from others so in some ways it prepares you to enter society because you don't have to have experience directly you can experience indirectly thousands of relationships going back hundreds of years and have a whole generative model of the human condition from that early reading so part of it is can you teach things like counter counter factuals for example can you teach your your child take hamlet put him in waiting for god all that god will put him in hamlet how does it change the story yeah i think that's absolutely right and and he started off that thesis with the notion of scripts um i think script is you know a very nice way of of thinking about the different kinds of models that are going to be plausible going to be allowable um um when self modeling you know when you know because the most important thing is basically a model of the world a model of the live world is essentially a model of me in a particular world so you know most of the most of the pressure on early learning is just finding out you know um first of all am i separate from my if i am what bits of the world can i move is it you know which bits of my so distinguishing between my body and non-body um and clearly as one gets more sophisticated more skilled then one develops quite quite a um sophisticated and deep literally with hierarchical depth generative model of me but of course that entails the kinds of things that i do and the kinds of things in this world in this culture and things like me do so i think that's you know um one way of looking at scripts and that has to be learned you know and as you i think you're you're also highlighting using culture using culturalism that brings with these deontic cues these these counter factuals that say well you can be that kind of person or that kind of person or possibly more relevant in this setting you'll be that kind of person in that setting you'll be this kind of person um just to give us to scaffold and to provide priors your empirical prize we call them mathematically prized that are like informed by experience and the structure learning and evidence accumulation that um the shape the structures that you will entertain um and of course you know if we are um 99 of our time is is is basically spent modeling the embodied me in some transaction or interaction with the embodied you then um a lot of these scripts are going to pertain to the kinds of things the

behaviors the social norms that would apply in that context i do emphasize the context you know i think we can be different people in different contexts we can contextualize our models um sure in the same way that if i was bilingual i would contextualize my narrative or the spoken narrative and given you know what i thought you would understand and i would understand stem from uh from you and having that context sensitivity that underwrites the generalization of one model for multiple contexts is this context sensitivity so the greater the latitude or scripts you have in front of you i think the more apt you will be to cope with any new context in later life uh you know i i think stories you know even down to the level of of of reading fairy tales with your bedtime yeah some of the some of the sort of um most um caricatured but but enduring shakespearean um um memes or structures or scripts are all there you know i'm i'm i'm a fairy princess so i'm some sort of all the grim fairy tales for example they have that sort of life structure in them that i would imagine most people would be able to identify with one character from a fairy story that you know five years later will be one character in the in their favorite eni blyton book or uh you know whatever children that read nowadays through to uh you know the political and other uh dramas that we read and read in adolescence so i think that's absolutely right and probably um you know there must be a vast literature in terms of evo diva the sort of combination of um development psychology developmental psychology and evolution so this notion that your evolution is just not the selective pressure acting upon a person but the person builds uh is in a particular environmental eco niche that is you know again massively and deeply encultured um and because we if you like shape that culture and that eco niche that encultured eco niche um then um the economy itself now becomes subject to selective pressure at another level so there's a sort of nice dance between the environment learning about the denizens and the phenotypes in that environment and learning about you know learning the culture of this environment and they both have to be like be optimal hand in hand as as we go forward one of the things uh continuing on with the uh the idea of scripts is that is reading uh losing its grip losing its grip and culture and society as one of the main forces to learn the that are the models that are generated in order to understand uh a theory of mind of others to understand yourself and understand the i mean what in your view because a lot of people now are reading less they're they're getting their information from uh tv programs i mean that's been going on for 50 years but with the internet there's a lot more distractions so that the kind of scripts that may be uh taught these days are short scripts as opposed to book length scripts that go into some degree of complexity i just wondered if you could react to that yeah i can but i i get to react as a pure

layman i have no professional insights here i do know people um um you know who make this their life work um and i think it's a fascinating area but you know i i'm just going to say that's a you know an incredibly engaging point um especially with the advent of social media and the you know the increasing um potency of adverts mini films within within 10 seconds now you're designed to provide extremely information rich uncertainty resolving material precisely tuned for brains that can only attend for 10 seconds or 30 seconds even to you know um synchronizing the um the cuts and the editing which with the natural tempo of our eye movements the way that we visually palpate the world so so will that have any material effect on the the temporal depth of the narratives and the scripts that we sort of start to inscribe in our generative models of you you're um the self of me of different kinds of me that that could possibly be um i i i again as a layman i i i would be surprised if if there's any real damage being done to be quite honest you know um you know there's long stories with with the long term new months are still intensely attractive from an epistemic point of view and and in terms of identifying with characters and the like um so there is there is ample evidence people still want the long stories um unfortunately it may not be and you'll have to ask me why i say unfortunate um but um unfortunately it might not be the the textual story uh but certainly if you look at things like game of thrones or your french police dramas or season one of you know the next hbo um you know mega mega drama series these these are structured long narratives um which people um which people are still um find um unavoidable and you know and will commit to so you know in terms of of um there being a sort of sea change in um you know short sharp fast trivial not trivial but um you know fast narratives with early closure scripts with early closure i'm sure that there is there are lots of opportunities indulge in them but on the other hand there's plenty of evidence i think that what people actually commit to when they settle down especially during the coronavirus pandemic um you know to to do that to go through their box sets the box sets are i think an epitomization of the long stories and of course the best box sets are those that translate in an easily digestible form the best books so you know yes more optimistic than one might think from a lay perspective well i i i think i i appreciate you sharing that i'm just curious as to whether it might have implications for your own mapping pro program and your own notion of generative models because the way we generate models depends on an input and if the input is a book you're going to generate one class of models if you're playing say computer games which uh i don't have anything against i think it's actually those they're fun but you're generating a different kind of model making about the world and those those do make a difference because as you say communication is a

shared activity and they're in a reading culture you share the activity although you read alone you share those books intellectually with those around you in your culture if that culture starts to shift then you're sharing them with a smaller group of people who model according to books as opposed to say playing computer games yeah i think that's an excellent point um and you know one aspect of that point um brings us to um an important sort of paradigm shift um at the end of the 20th century which was towards sort of an embodied perspective an inactive perspective a situated perspective on sentient behavior which tries not to deny the key role of the physical way in which we sample our information get our information from the world and indeed you know on some radical accounts the very physical and embodied act of sensory you know soliciting sensory information from the world could be read as as perception so i think one one attractive example of that is the the finger fluency that teenagers now in fact millennials now have in terms of being able to type on on on an iphone now i don't have an iphone so i just don't have this fluency but they they will have a substantial part of their motor and pre-motor and supplementary motor area devoted to this fluency which um you know that would parallel um parts of my brokers area problem zero 44 that i devote to using more my favorite words you know when writing them down so there will be physical differences in our motor compensat our communicative acts that will be written into and entailed by different brain structures and different fluencies so i think that's absolutely right um you know so the new authors may may not now be people who are writing books they may be people who are engineering games yeah sure and so there's more money in that than there is writing books um i can attest to that well if you just want the money then well that may or may not be a good thing uh you know it um the question is that a good thing or or or a bad thing from an evolutionary perspective you know you know i make no judgments but it certainly is a reality um you know that you could reverse engineer if you gave me a young person's brain from 2015 relative to a young person's brain in 1915 there will be structural differences and if i understand the nature of their genitive models entailed by their brains i will be able to say that this is the kind of uh you know youngster who um who reads books and uh you know um and i will be surprised to see um gross differences between you know all of their friends uh you know in their conspiracy or their in group whereas the millennial um subject would have a different kind of brain who would have different kinds of fluencies and you'll be able to see well this is the kind of new reader who probably does engage in narratives by watching um netflix or playing games online with their with their you know with their friends and siblings so as a neuroscience you could see uh same territory but different maps yeah yeah yeah yeah

the way it comes you can see the the mapping uh has been of a different kind and different quality but the territory is recognizable but the map of the various trails and where you're going is take some use to getting used to because you don't have that part of your brain which has looked at that kind of map before yeah you don't have that sort of physical language you can't articulate and if you can't articulate if you can't do then you can't see you can't understand uh you know so so you know the essence of um understanding others um is this the notion of this shared narrative which means that if i see you doing something then i can use my models of the kinds of things that i do to infer what you intend and so if i if i can't do something um then i can't understand i can't make sense of um of you if you did the thing that i cannot do um so i you know that's not a sort of mystical statement you know it really really does um bite hard say neurology where people with parkinson's disease may have difficulties adopting certain facial expressions or responding swiftly in terms of initiating certain kinds of movements which means that they can't do it but crucially they can't recognize those kinds of actions as commutative acts in other people so you just lose the language so i could never in any true way understand what a finger thumb flip flicking duster on the tube is is trying to do because i just don't have that as part of my generative model so i can't model them they are another species for me i mean not in any malignant way but you know they are now a different generation and i am not part of that generation there will always be a communicative barrier uh which is not a bad thing but there will always be that be that barrier simply because they have a different way of expressing i think it comes to the notion which um you you you brought to the table around about the modalities you know the modality um does um entail the kind the physical ways in which we communicate in which we which we share narratives um and you know if i don't have i don't speak that language or i am not able to uh play a particular computer game or i am not a subscriber to netflix there is going to be a you know a fundamental um disaggregation segregation uh of of different cultural groups um being generational or or clinical literally no longer speaking the same language which is often the way a generational divide is is put there's a a breakdown of communication because the medium in which communication is used is it's just different enough it's just like if you hear a dialect uh even in your own native tongue you still have to strain a bit you may pick out some words but there'll be other words which are not familiar to you if you go to edinburgh you're going to find maybe in a pub there that there's someone you're missing some words uh which are common to everyone there but are not common for you yes no absolutely yeah and you know the dialect is very interesting because you know that introduces a degree of uncertainty

so if you just think about what things are we attracted to and the valence of things and if the key imperative is indeed this um imperative to resolve uncertainty then the uncertainty induced by what did she say you know right it starts to become slightly aversive and and and it is that kind of noisiness that that uncertainty which you which irreducible which you will ignore and you'll go to another more reduceable kind of yes this is my kind of book this is my kind of story this is you know this this is this is a tv series for my age group because i understand what's going on and at the end i will get that new month i will resolve that uncertainty but if you infer you never you know for example um you know new kinds of uh popular music that just do not you don't have a genetic model for you can't predict what's going to happen next you'll actually kind of find them quite immersive because they do not afford the opportunity to resolve uncertainty to improve your model of the world um you know in any meaningful way uh and you you you'll you'll you'll you'll stick to as you get older you will certainly stick to the kinds of uh sources of information that do have that epistemic affordance so i mean there there is kind of an emotional uh homostasis that i think that people naturally drift too so if there's something that is incoherent not particularly understandable or pleasant they either avert their eyes or pretend that they didn't see it this this is part of human nature of what you choose to see and what you choose not to see it goes back to john burger ways of seeing and again i think that comes from from reading which i think is a segue in let's go take a time machine back to york you're now about 10 years old none of the medals none of the accolades or the fellowships are there you're a boy reading a book tell me a little bit about the that environment in which you are reading a book um it's an environment that is shaped very much by my family particularly my my parents really so i think to to understand um that um and the books that i would pick up and read um um you had to understand the why those books were there so my mother was was was um a nurse um um and when she got married it was a in the age where um it was you know the scripts were that you sort of invested in in your in your children as a mother and the father went out to work my father was a civil engineer um which meant we had to move around the country very much like an army child to wear the new structures he was actually a motorway bridge engineer so we had to move around the country and he spent a lot of time on site so i had this curious mixture of an extremely bright man um with a passion for um physics and engineering and an extremely bright mother whose only focus at that time was how to bring up children properly and so she comments lucky loads of um books on um psychology popular psychology uh um for for her edification and um enrichment that was nice a nice compliment to my father's books which

were largely all engineering and his favorite book uh was space time and gravitation and i and i by sir arthur eddington um and i only found out about four years ago um how favorite that book was because apparently when trying to court my mother when they're in their teenage years he'd made her read it as well that had to be true love [Laughter] because it's the only book on your list you have seven books i i i've never heard of this book so i did a little bit of research the the most recent review i could find was in the bulletin of american mathematical society january 1921 and the only place i could find to buy a copy was walmart canada's website for 89.40 it also said the other two pieces of information it had no reviews and two it was almost sold out so please tell me what age you read this book your mother read it as a teenager how old were you when you read this book i'm i'm it would have been i must have been about um 10 or 11 years of age several times um um but it was it was it was um at a time where we're actually having maths left lessons at school so so the narrative the language of maths was starting to to emerge um and um so so i'm surprised there were no reviews of it because it was um you know i think latterly it's seen as a classic in the sense that it was the only accessible way that an informed sort of early 20th century layman could get a good handle on general relativity and einstein's notions so sir arthur eddington the the author um was was famed really from for being able to translate einstein's ideas into a way that um that enabled people like you and me to you know to grasp the essence of it um and it's still i think a very readable um readable book if you want to understand the uh you know not only what is relativity but uh general relativity but also um you know how you might take those baby steps conceptually and possibly how einstein and people at that time were thinking and in thinking out of the box and you logically arguing themselves to the way that this world must be so coming back to some generative models of a very grand sort here certainly in the context of physics so so you know that that was my that was probably my favorite um favorite book because you know it really did have that epistemic affordance that we were talking about it really did um not only engender a whole space of counterfactuals about the way that the physical universe could uh could work but also provided a denumont in terms of resolution of uncertainty so the big thing about the story that that book tells which is basically a story of um conceptual exploration in terms of the you know the the way that the physical universe might might be structured but it was written by somebody who was actively involved in securing the evidence for it um his 1919 expedition to off the coast of brazil for the eclipse absolutely so i mean it was a great you know in terms of science history it must be such an exciting time for scientists who also were explorers and adventurers it was an adventure story as well as an

exploration of science is that what really kind of hooked you on science uh at 10 years old was this book yeah i think so i mean i think as as with most um people who subsequently became natural philosophers or natural scientists uh you know in the victorian sense um those opportunities to find out how things worked um again had this epistemic affordance that was irresistible and that particular book just took you to the edge um and you know if you ever have the chance to read it you you will see that it actually starts off with exactly um something that we we've already focused on as a fundamental way of engaging with the world which is questions and answers yeah if the purpose if we are here in virtue of being curious creatures um then what does that mean it means that we basically seek answers and how do we do that in an informed and efficient way we have to ask the right kinds of questions so questions and answers um you know are central to model building and then living in the and interestingly space time and gravitation starts off with a conversation that is just a series of questions and answers between behind you know from and between three protagonists with slightly different perspectives on the way the world works you know mathematician and the physicist and so on and so he actually starts with with just a conversation uh so you can and that's what i meant by it was a fascinating insight into the way that people were asking their questions and offering answers some correct some incorrect um at that time but it really pulls you in as well because you will be wanting to ask that question and you want to hear what the answer was so it was very compelling reading just the first chapter you know and it's a device that i subsequently used uh when writing with friends in uh you know in philosophy proper and that's the best way to get to the heart of the matter you know to sort of get into the narrative through a series of questions and answer but then the adventure as you said the the practical adventure of going off and doing these experiments you know uh and the history of experiments that underwrite this you know this particular view of the world which was very different i mean that's something very refreshing about about you know the way that you know our sir arthur eddington paints a picture of the universe where they have different shapes you know well beyond does the universe have an edge uh you know what's what shape is it is it hyperbolic there's a notion of it at the end of the edge of the universe even exist does it close back on itself like a big circle what about yeah and it was just so fascinating just again a little bit like the other kind of actuals that we were talking about before the different ways that it could be which you you know you'd never imagined but now you had and once you'd had that experience of imagining another another way of looking at things then that made it easier just to entertain other kinds of counterfactuals and of course that's the mark of a good scientist a

counterfactual is just a hypothesis the mark of a good scientist and natural philosopher is asking the right questions by bringing counterfactuals or alternative hypotheses to the table and that takes and from imagination you're going to have creativity coming out of that because the more you can imagine the more you can create and construct and different models so you're learning this at 10 years old uh a book which was read by both of your parents that actually was a condition for the father's decision to want to marry the mother and she went along with this so did this become something you talked about in the family did it become kind of a family uh touchdown the space and time and gravitation i mean it has everything all the questioning and answers the process of inquiring the world i mean did these things come up at home not at all no i had to my mother was completely non-mathematical would have told my father off if he introduced that in family conversation so no no this is all under under the cover it was all implicit um you know my my father um sort of you know was a was a great educator but but in the best way possible in terms of just providing learning opportunities so he gave me a slide rule and said how does that work and never spoke about it again but i spent months working out how a slide rule worked at that age so before i ever met the concept of a logarithm in mathematics i already had a fundamental understanding of you know arithmetic and geometric transformations and the role of logarithms simply be you know trying to appease my father because he asked me a question i never told him the answer and i think that space-time gravitation was one of those devices he said read that yeah it's like a slight ruler but i do know i did notice it was interesting you managed to trace down a copy although we never discussed it we do discuss things now you know uh you know now he's retired but not in not in those days but it is interesting that after i had read it um you know several times over the years before i left home he made sure he got the book back and it's his he's coming down in his favorite bookcase and he won't let me have it until until he passes you know for him it was a very important book as well but but it was all done silently no my mother would not have um find those kinds of counterfactuals at all entertaining i repeat she was much more into psychology and ways that people thought and think you know so she was much more verbal in her analysis um so so there wasn't discussion of general relativity around the table right now i i've heard you say as well that underlying everything the the foundation is mathematical uh that is is the place that you start and then you go up from there to explain the world and i'm wondering again if this book space and time and was one of the early influences for that view of reality of model making starting with a mathematical equation yeah no i i i think that's absolutely true i mean it's both true that i um have said and believe that um mathematics i

think is the most efficacious language um the best shared uh narrative that we have um um that you know that transcends most other spoken languages or written languages um and you know it's unreasonable efficacy boils down to its simplicity and it it's delicate but but um beautiful structure um so i think you're a vehicle as a way of talking about ideas and that ultimately has to be reduced to the formalism of mathematics before but i think anybody else will understand it ultimately i'm not about now but you know in five or ten hundred years time all the great advances i think um um you have have or um can be articulated mathematically i'm just catching myself there because of course you know you might argue that natural selection um was a great biological idea that wasn't necessarily mathematically couched but i think in terms of making um natural selection work then you would um you would certainly then formalize it in terms of genetic algorithms and you know you know pricing you know sort of uh you know fundamental equations of evolution and the like which actually are very close to the same equations that underwrite the free energy principle in terms of us as curious creatures making inferences about the world right we are inference making creatures for sure let's turn to the first of the fiction choices uh one of my favorite and i think favorite of many people 1984 by george orwell uh what age were you when you first read 1984 oh i i can't remember with any veracity um i must have been i'm just imagining where it was it was on the bookcase at the bottom of the stairs where my mother kept her favorite books um and in virtue of being in that house at that time i must have been about about about 12 yeah 12 to 13 yeah tomorrow okay so you pull it off your mother's uh book shelf was this a book that you just devoured over a day or two or you lingered over it for a longer period of time i think i probably devoured it i got i'm quite a slow reader so now i'm in fact you know i have to confess since going to university and since you know committing to a rather hectic work orientated career i don't read the pleasure anymore because i have to review scientific papers um and the occasional book nearly every day so i just have no space for for pleasure reading so but in those days um you know you had the time and the space to to do it for pleasure and i probably was quite voracious in going through it so you know i my mother had a collection of penguin classics so i just went from right to left i remember that i do remember 1984 as being particularly uh particularly compelling uh there were you know there were other things that i'll just have to say in the doors of perception but 1984 was you know it was quite enchanting and frightening yeah so i do i do remember that one quite acutely so i have gone through it quite quickly um you know late into the night under the covers uh kind of uh stuff what was what did you find at 12 years old that was the most compelling part of 1984 oh it is a long time ago i'm

remembering this like you might remember your a dream and not very well so we're talking over half a century since since i actually read it um but the kind of the feeling that you had can you remember that feeling that experience yes it was just the horror of realizing that my comfort little world shouldn't be taken for granted um you know when you know when the veil was slowly lifted on the kind of world that the protagonist um in 1984 had to contend with um it was you know an eye-opener that such societies uh you know uh such um cultures could actually exist um and may actually exist in other parts of the world i didn't know at that age so it was just a revelation that there were other universes universes of a cultural sort of an attitudinal sword are you going back to these these sort of the norms that um you know your how should i behave in this context in this in this society so that was that was an eye-opener and i and you know curiously attractive because it was just so frightening uh and you know once you developed this hunger for uh another kind of culture another kind of society and what you couldn't couldn't do in that society and all sorts of things which must resonate with everybody you know you know all the little phrases like room 101 double fishing all all of those classic things which you know whether or not you like the word meme i think the meme applies to to these things that once you've seen them uh they stay with you for for life and so you know these these were you know every few pages was an eye-opener and of course there was always this i can't know what this is true or a false or made memory now but there's always this sort of notion that he could get out of the city and you know and go off to these these green fields you know and escape with his with his girlfriend at some point so i love that as well and i think that's possibly because um you know the way i was brought up in yorkshire there were two bits of my life there was sort of hard work sometimes actually getting used to being in a new school because moving around a lot you know settling into a new uh new environment and then holidays off on the yorkshire moors so i've always had this poncho this this um always light reading about um things like so far from the mudding crowd by getting out there um back to a quieter pace of life and i i just remember that there was this sort um this was this escape for the protagonist in 1984 if he could get there if he could realize it but it was clearly never going to happen in any meaningful way that was that was almost again quite quite attractive yeah it was not the hobbits in that case you know this this was a pretty dark world in uh that's portrayed in 1984 and what what's interesting is it's a book that has endured for uh 60 70 years now i think it's about 70 years and it's now out of copyright 1984. but let me just give you a couple of phrases here from from the book the party told you to reject the evidence of your eyes and ears it was their final most essential command in other words your

ability to do what you're talking about in your your paper of asking questions asking better questions resolving the uncertainty you no longer have the autonomy to do that that has been stripped from you questions and answers are over this is the script the only script there are no questions and we may change the script tomorrow we may be at war against these days or tomorrow we may not we'll let you know that's that's the world of 1984 and part of the horror of that is another of the quotes not merely the validity or experience but the very essence of external reality was passively denied by their partnership there was there's no generative uh models uh that are done independently uh in 1984. uh it is a chilling book uh and it's a book that fits into some of your other early fiction as well it's it's a statement about power and those who see seize power never voluntarily relinquish their power those were great excerpts i've forgotten them about it's all coming back now yes so i was wondering whether you were going to um talk about the ghosts of that withdrawal of our life affirming and i would say mathematically necessary um um quest for evidence for our own existence in you know um uh whether whether that was being recapitulated in the in in the modern era by things like fake news you know in a sense you can imagine fake news is basically saying the truth does not matter obama truth decay there is no evidence of any worth out there anymore i think that's equally frightening uh but you know but painted in black and white uh by all well i think that that's that's imaginative well again let me give you a couple more uh quotes uh to refresh your recollection he's talking about the ideals set up by the party was something huge terrible glittering a nation of warriors and fanatics marching forward in perfect unity all thinking the same thoughts and shouting the same slogans perpetually working fighting persecuting 300 million people all with the same face and then power is carrying the human mind to pieces and putting them together again in new shapes of your choosing again i ca i could perhaps see a 12 year old uh are all reading this and some of this it seems to me now is reflected in your own research and your your own modeling is it is a way to showcase what the human brain mapping should be it should be have the freedom the basic freedom to generate models of without manipulation without interference that there should be some choice in this and there's george winston with his girlfriend julia trying to find some small way to model a traditional relationship and being unable to do that being bounded in in such a way that the only thing they could do was to submit they could no longer question they could no longer answer but they love big brother so i think with 1984 we find a book where again curiosity comes up in orwell's world there is no curiosity that's allowed there's no creativity there's no and so creativity and imagination go across just language you go across music it goes goes across the

mathematics that you use to describe the universe because it may disable and disable institutions that have a vested view and what the model of the universe is copernicus this is you had your copernicus moment when you were talking earlier about suddenly i understood growing up in new york that york was not the center of the universe there were things going on in other places that were quite terrifying the way to interrogate the was not universal it was local yes sir absolutely and perhaps that was i mean you know looking back clearly you can see that kind of utopia is just not sustainable it's everything sure so what one way that people um philosophers describe this sort of um the ability to um pursue life as a curious creature is self-evidencing the you know the existence is just acquiring evidence asking questions and acquiring the evidence for your own if you're in so you're just mulling over what the the excerpts that you've you've just um shared with us what that is describing is it's completely unsustainable and it's frightening because it cannot exist or if it does exist it won't exist for very long it is the exact opposite of self-evidencing that latitude that creativity that opportunity to ask questions to reaffirm and then change my mind about my place in in this world so perhaps that's why it was curiously frightening but as you say very dark yeah it's a system but you a lot of what orwell wrote about in in the 40s we can see playing out in various countries today where there are more totalitarian measures authoritarian measures to limit free expression which again when whenever i hear limiting free expression it's limiting the right to question question is the challenge the challenge is is the model may not be right it needs improvement we need better and more detailed answers and that requires being able to say the current model is flawed and so you have millions and hundreds of millions of people who don't want the questioning who are quite happy to have the party decide what the truth is and that everything else is false are they happy good question good question whether that brings some kind of emotional resolution maybe it is at just an emotional level that happiness for them is is having the security that there is a community of who have faith in a particular stagnant modeling of the world that the of the world of course probably never existed because these are fantasy models but they believe them to be real it's not a simile or a metaphor it was like nope it was like napoleon saying i am caesar not like i am like caesar i am caesar so there there is that kind of internalization that can go on and seems to be going on yes yes you're attempting us be into a political discussion of contemporary no no i don't i don't want to go there okay but but you know we we may uh go there a little bit in terms of the next uh book which is quite political homage to catalonia which is another george orwell uh this is non-fiction as opposed to fiction but it is a absolutely memorable i remember reading this very

young as well so i'd be very interested how old you were when you read this and what what kind of influence it had on you at the time and now so clearly my mother put all her george l or el um or well penguin classics in order so it would have been red next to 1984 and you know that one was in a similar spirit to um just realizing the world could be in a different uh you know in a different way or there weren't um there were realizable universes out so you should never take your own world for granted what i took from the homage to catalonia was it was the autobiographical aspect i didn't um until you reminded me remember it as an autobiography i thought it was just a story about a young man's adventures watching the you know a war in in you know originally things it started in barcelona but but the the other thing for me was um i could see me in 10 years time and that there were different possibilities about the way you used your life um because i had assumed that i was going to be a good boy you know probably thought at that time that i was going to be a psychologist because of my father's passion for mathematics so you know i probably already had committed to a life as a mathematical psychologist or like some kind of scientist involved involved in the brain but you know i was impressed by orwell's sort of moral and political convictions you didn't have to do that you could just say no i'm leaving this country and i'm going to go so just a different way of scripting was a real thing and it actually happened and here was this elegant young man talking about how you know how he chose to to spend and risk his life um you know at a time you know when we well i was at school i have to make choices about exams and the kind of education and the career path that you you that you were likely to pursue and clearly those sorts of questions really did not bother george orwell you know he was out there doing doing the right thing uh you know having adventures he left eaton and went to spain to join the uh partisans there i mean one of things i love about orwell and it comes out in homage is his humility is beware of my partisanship my mistakes are fact and the distortion is inevitably caused by having seen only one corner of events there's there's a beautiful sense of humbleness in that statement because you know anyone who's been in a situation uh like that knows that what you're seeing is not the whole context things that are going on or orthogonal to you are multiple you have no knowledge of them and so what you're what you're reporting on is what you've experienced and what you've seen and there are very few people and orwell was one of them who was able to acknowledge i could only tell you what i experienced what i saw and i can also tell you beware of the propaganda from abroad written by people who have never been here who have never been in battle have never seen the things that go on because they have an agenda yeah one one of the things that uh is interesting about homage as well is

his humor arises i love this quote for some reason all the best matadors are fascist i i think you know i've been thinking about that maybe it's it's the public display of the power to slaughter the bull in front of uh a stadium filled full of that would be attractive uh there is a certain level of fascism in the matadors role and performance a wonderful play of language and the thing with thorwell is he's able to layer his observations with these kind of deep insights by just the right metaphor at the right time so orwell also uh in addition to his reporting on the war had things to say about the uh the ruling class in in england i'm just wondering as a young boy growing up if that uh had any impact at all here's a quote on class whether the british berlin class or wicked or merely stupid is one of the most difficult questions of our time and at certain moments a very important i think it did it did influence me um i mean it's difficult to say whether those books influenced me or whether i have chosen those books because i have i have subsequently um adopted a particular political and view um but i suspect the former um you know interestingly your point about your beware um the um being partisan on the basis of of you know the secluded amount of of evidence that you gather from from um you know from your exposure to the world having you know in contrast um i probably inherited a lot of my political views not obviously from my parents yeah but also by just assuming that orwell was right so you know i subsequently grew up to be effectively a socialist uh of an understud understated sort but if i had to commit to one side of the political divide or another i'd certainly be um in in almost a direction uh simply because i i i took him at face value because of the honesty and the integrity of his writing um both the fictional writing but also the you know you know what struck me you know him writing about you know his own experiences as a you know as a young man yeah i mean you're absolutely right i mean that autobiography uh is an excellent example of honesty brutal honesty and you know he talks about when he was shot he says two things went through his mind first his poor wife what was she going to do her husband's killed now in spain he says my second thought was this is just crazy this is stupid this is some random shot of i'm standing in a trench i'm not being heroic or anything i'm just shot talking to someone it is absolutely beyond my comprehension that i'm going to die in this stupid pointless senseless way and all the nurses and all the doctors said the same thing anyone shot through the throat they always die you didn't die you are the luckiest man alive so there's orwell saying you know what there is an amongst the absurd there's also random luck i had both that day the absurdity of being shot in the place i was shot at the time uh along with the fact that i survived when i shouldn't have survived i forgot about that but now you've reminded me yes that was a very poignant part of yes thank you again i think going back to

generating models is and again it's generating models and uncertainty go together in in this way is that there is a degree of randomness and chaos and luck that's involved and you find that some reading as well that basically a character can have some either good luck or some bad things are happening outside of his or her control and whether homage to catalonia is one of those books i think there's there's something to be said that there is a certain degree of random spin the wheel of you end up in spain as opposed to portugal or thailand or wherever and things happen there that would not otherwise have happened to you if you'd gone another way it's the counterfactual again so what we're learning then from your reading list is there are a lot of counterfactuals that the characters and the authors are yeah from eddington to to orwell so maybe there's a lesson here that delicious diversity yes the diversity the the counterfactuals continue to proliferate if you allow them inside of a creative imagination found in the right series of books which leads us to lateral thinking which is a non-fiction book by edward de bono what what was your age approximately about 30 um so this was the the um this is one of my mother's books you know a whole series of popular psychology books but i remember that one in particular um simply because it drilled down possibly in the sort of popular psychology fashion um on how we thought and in fact it actually addresses exactly the other sort of the model building the creativity um so the natural thinking you know could be read as just thinking outside the box gaining access in your head to that diversity those counterfactuals you know that sort of enable you to render your understanding your narratives your model of the world generalizable and context sensitive so i i can't remember very much about it but i do remember being intrigued by the importance of um letting go of preconceptions and um celebrating the brain as literally a fantastic organ a generator of fantasies and hypotheses counterfactuals to to enjoy itself in thinking about different ways of solving problems or explaining explaining this and that and it's certainly something which um you know became a scientific question in later life you know thinking about how how on earth do you get a computer to do this if you wanted to simulate aha moments of insight in silico you know what your how would you engineer that what you know what are the principles that underwrite that kind of that particular sort of creativity and and resolution so you know sometimes it's it's the creativity is almost paradoxically finding simple solutions you know past emotions explanations oh it's just like deflationary uh your explanations so you know i got a sense of that from from what i remember of um sure of developers edward natural thinking for me i read it quite young as well is it some ways it was kind of a simulation program of surprises of how to surprise yourself and in some ways it's like test crashing a car to see what gets bent see

what works what does it how you can make it a little bit stronger and better and so forth and that lateral thinking was that way testing the model that you have and you test it in hard ways that surprise it because you know no model has all the that at some point there's going to be model failure and what you're looking for is how best and most efficiently to get to failure of a juncture and to learn from that and by improving the model go back through the process and it's in basically an indefinite process because the model never gets totally perfected yes that's that's one of the things i think i took away from that book when i'm well that that's it i i you know i i've forgotten that aspect but you're absolutely right and of course that is exactly the structure learning we were talking about before you know testing against the evidence at hand breaking the model reassembling it adding little bits in or taking stuff away that is exactly the sort of you know the um the the thing that underwrites how we learn to live and live in this world uh so so that so technically that would also be known as bayesian model selection where those hard tests of your hypothesis your explanation are essentially seeking out data that would disambiguate uh between the hypothesis of the right model this is the wrong model right so you know good informal formative data is often the data the evidence that destroys your hypothesis breaks your model that road tests your car in a destructive way then you know that was wrong and then you move on to something which is you know so that i hadn't seen got that perspective i haven't remembered that perspective but it's it he does feel absolutely right you know from a first principal account of sentient behavior you know because in some ways it i mean de bono's book on lateral thinking kind of underlies the process of scientific thinking which sets it apart and maybe it's that kind of process that's difficult to communicate to a non-scientific audience because to think there are people who are trying to break their models in a world where most people are trying to protect their models at all costs you see that they're talking cross-purposes yes that's a good point yeah um so you know you know if i was lecturing students in psychology or physics i mean you know the whole point of a good experiment is it provides definitive evidence for or against your hypothesis so good data is data that will actually enable you to move on from a particular hypothesis so there's always there's always a counterfactual in science is the null hypothesis and the alternate hypothesis and what you want and those data that allows you to disambiguate if you get some data you left 50 50 you that's a bad scientist again it's just another way of saying asking the right questions so what is a question it's just a move on the world to solicit some data which resolves your you know i think that's a lovely description of the scientific process uh which you know begs the question um you know what what what instances of people trying to

protect their model or their ideology um you know is is that part of this sort of uh confirmation bias and bay's optimal maintenance of a particular generative model maybe you know you know if you can find the right environment in the right and the environment or the culture permissively provides the right evidence perhaps that's a viable solution as well but of course then you take away this context sensitivity this counterfactual richness that we were talking about before you can't generalize you know that 1984 utopia will work provided everybody commits to that and there are no external perturbations there's nothing else in the universe that's going to come along and undermine it or recontextualize it i think and again uh your markov blanket is a useful metaphor in this case where if you have one that is so thick it's impenetrable it can't be changed it's not subject to question and it is the titanic it will never sink uh as opposed to i think the scientific process markov blankets are like the universe it can be in multiple shapes and configurations and some we learned from better explanations are better shapes than others but no one has the right to say one is the preferred the only shape yes no absolutely yeah so i was just thinking about um a very thick mark off blanket i'd never heard that before but it's a lovely idea which of course would would render you completely insular you know are you isolated yes you're very thick and you're immutable unchangeable um and provided that thick blanket protected you from the rest of the that will be fine but of course the universe is going to be around that blanket at some point yeah then you get this sort of you know this dynamic so it's a very good point from the point of view of physics because the whole point of the markov blanket is that we're not islands you know we have to exchange with the world outside our heads the world outside our communities and if you're a cell the world outside your cell boundaries um we are open systems and that does require exactly this sort of transformation and transmutation and dynamics and reshaping of the markov blankets that separate me from not me or you if i'm not you it is it did occur to me in your uh earlier conversation with me that there might be a range of these markov blankets and that there may be an element of the population where that that blanket is so overwhelming it can't be changed in the other end it's so fragmentary it is easily destroyed either of those are not going to be reproductive of a happy life of a life that's rich with texture and able to accommodate surprise and uncertainty yeah no that's absolutely right i mean you can see examples of of this all around you you know the mark of blankets that are so friable so labile so capricious that they only exist fleetingly um is a very different definition of things that dissipate die and decay you know so all of those d words which have negative connotations are just descriptions of the dissipation of the markov blanket a thing losing its thinness uh you know as a markov blanket

disappears so stuff that doesn't exist or decays or dies or dissipates very quickly that's one end stuff at the other end and can exist but it exists like a stone or a diamond for as long as it is allowed to exist and does absolutely nothing and it's the goldilocks regime in the middle where we live which is not a stone uh but neither is it neither is it a forest fire yeah let's go to the lord of the rings uh it's interesting you mentioned you know growing up in uh the countryside uh in york and that you know uh as you probably know j.r.r tolkien grew up in countryside in worcestershire outside of birmingham and in some ways the lord of the rings is his um perception of that childhood of that place of those people of those times the culture the myths and he brought in all of the myths with the slavic finnish celtic greek norwegian he was a great linguist at oxford university uh we had the chair there and these books i think had a profound influence on uh multiple generations i know i'm not certain if they're read today they than they would have been in the past but i'm very pleased to see them on your your list because uh again we're back to 1984 in a way we're looking at symbols of power sources of power the most powerful people setting the communication moral agenda of what people can do what they can think and that the power in this case is one ring by someone who is power hungry sardone uh living in his castle in moldova and we have a small group from uh the forest the hobbits frodo and sam who venture out on a reverse quest which is a counterfactual if you like it's this is not odessa's this is the opposite they are at home unlike ulysses who was trying to get home so being at home they needed a reason to leave and the reason to leave was to take the ring that bilbo had given photo his cousin to go back it was only one place where it could be destroyed which was the center of this power and the whole story is the journey back to mortal in order to destroy the ring that threatens to destroy middle earth which is where the hobbits live so how old were you was this on your mother's bookshelf as well it wouldn't have been amongst the psychology books but where was it this was this is my book so and and in a different way from 1984 you know i think this is my first exposure to the um the fantasy literature um another world but so this was much more escapism it wasn't about having your eyes opened i recognize everything uh that you know tolkien was writing about whether it was the countryside an idealized country your countryside you know homestead um or the mountains that i was familiar with with the pennines indeed with the occasional trips to the welsh mountains or the building um or the scottish mountains um so and also i i can't remember why but i i also loved sort of um old english folklore you know the details of of of sir arthur and merlin and the like and you know if i could have been one person at that age i would like to be merlin as a teenager so but so that all of these sort of um notions and and

narratives and the atmosphere um and the you know the compass of this new world that that tolkien had created was fit very very comfortably with all the things that i that i love but it was somewhere you could escape too and that had all the right ingredients you know the the the one ring the simplicity the you know david and goliath uh kind of situation the uh overwhelming odds uh the heroic venture with a small fellowship and that within this fellowship they could find the resources creative resources in order to overcome what seemed to be impossible odds yes you know the hallmark of a great story will they do it will they succeed yeah where uh rather than the uh george winston ending in 1984 where he says i love big brother here the big brother is in for a fall and it i mean one of the things i think that's interesting about the the lord of the rings is the symbolism of going back in england to really pre-anglo-saxon england before you know certainly before uh with the normandy invasion that there was a sense of people living with nature as part of nature that really the the big changes came with the agricultural and industrial revolution and that that changed everything that took communities away from the force destroyed the forest and that was in a sense uh something i think the token regretted very much the the sense of if you want to just that was the power the new power was these new technologies that were coming in and basically uh changing everything and everyone so i there must have been changes that were going on when you were growing up in york as well uh in in the rural area you could probably see even in your childhood that uh things were not going to stay the same yes absolutely well i mean you know my father was instrumental in that in terms of building uh a lot of the infrastructure so certainly in england at that time that's when you know a lot of the uh the the technology and infrastructure which we you know which we currently enjoy was being assembled so things were changing and of course mixed with the slight um sadness that my father would have to leave the home to do that so i didn't necessarily see this infrastructure building this the mechanization um um of of england as as necessarily the best thing because it entailed actually the absence of my father said he's a younger man but you're absolutely right and you know another authorized love was thomas hardy because it just took you before this time it was just a much safer time it all struck me that the past is a much more comfortable pleasant place to be because there really is no uncertainty about the future because you know what happened um and certainly in this so pre-technological setting i've always been drawn to dramas in an escapism sense stories and and literature that is in a quieter gentler more contemplative time usually with lots of fields and grass and vegetation well the the thing about the past is that it's basically drained of most of the surprise because we know what happened

uh and what didn't happen and there can be revelations that come up here and there that change our idea of what happened and we may be a little bit surprised by it but being surprised about a new piece of information from the past is quite different than being surprised about what's happening today or what may happen tomorrow to prepare to prepare for i mean just just to perhaps put this in context at that time you know that there was end and the cold war and i remember you know having um government information films at school about what to do in the in the case of a nuclear explosion so there's a genuine fear that the world was going to certainly amongst my age group probably naively there was a genuine fear that the world was going to explode um and you know you know watching pictures of nuclear bombs go off induced the kind of deep existential fear and angst that you have during a nightmare so there was you know certainly in the era of thomas hardy in wessex or in um in middle earth you knew that couldn't be a nuclear bomb um but there were still great challenges ahead you know there was still sort of dramas and betrayals and you know so there was still a really important narrative but it was a it certainly did not entail the kind of devastation that industrialization and technology and certainly nuclear technology would entail if it all went uh went wrong right at that point in in sort of english history yeah it's it's interesting as well because i know you've written a lot about you know communication and the importance of that is that it's been said of token's uh trilogy that he's really writing about the loss of communication between man and nature that there were lines of communication and that it part of what he was trying to do was to write about that breakdown the relationship between man and nature and uh you know with climate change these days we can see that actually uh that breakdown has had quite significant consequences if we uh could move on to lord of the flies from lord of the rings to the lord of the flies i'm not certain if there's any perfect segue for those two titles but uh we're going to try our best were you living with your parents in york at the time you read this yes yeah about the same time um probably a little bit older because i also remember doing that at school or at least having having discussions at school so obviously i got to the stage of doing my studies for english literature but i had read it previously um and this was um this this was at the tail end of my mother's collection um but i think spoke very beautifully to psychology um and you know the way that we relate to each other um you know when circumstances when circumstances change so this this was you know um portended um a growing fascination with psychology and psychiatry and eventually i became a psychiatrist on route to um uh becoming becoming a theoretical uh clinical neuro biologist so this this was this was um an eye-opener um not in terms of the way that the the

world works but in terms of the way that we work so for me yeah again very very compelling read i mean you read this as an adolescent and really it's a book about adolescence without adult supervision of in a sense dividing into factions and the factions might be called one would be maybe ralph who was more uh of the for civilization and democracy and consent and uh rule of law and then of course there was jack who was just the opposite he was into hunting tactics power games domination uh piggy who was the intellect and the rationalist and simon who uh was the incarnation of good and holiness so we had all of the adolescent characteristics there uh put together and from from your observations as a psychiatrist in the past was it a reasonably uh valid look at how adolescents divide into factions and the kind of values that emerge from that yes i i think i think it is and beautifully painted you know there's different dimensions and aspects and also very apt descriptions of all my classmates the different classmates you could be and i have to say almost to you you had a choice you could become any one of those people in in different contexts so again you know different scripts for different ways of coping and coping here in a very changeable unconstrained context where you were in charge i mean it certainly um it certainly was a you know a detailed picture of but in many senses one could imagine uh grown-ups disintegrating in exactly the same way we are without those societal strat infrastructures and cultural constraints in play to keep us all together all you know on this you know reading from the same sheet or him sheet but certainly um you know as an adolescent um he was a good library or repertoire of different ways of behaving and i i can remember um yeah wanting to have mixtures of of of the you know the key characters you know i was growing up at that time um having moved because my father was now building motorways in north wales in england so moved to a uh a different school where there were um a lot of people um where the nearest city was liverpool um and some of them were were beautiful characters very you know sort of compassionate um some very religious some your hard people um sometimes obliged not to be piggy which was a natural tendency given my intellectual leanings so i deliberately tried to be uh unpiggy and learned it it took me it took me about six months to smoke a cigarette without being sick but i had to do it get that edge um so it was you know it was great to see that you know to see all these characters uh on different ways of behaving um uh you know play out but you know also how they all those trays become very much accentuated and caricatured in their expression when the breaks are off you know when the context and the constraints are not there you know all the paranoias come in and you know the outlandish surprising ways of responding um when you can't predict what is going to happen can sometimes lead to quite catastrophic outcomes

so that's that's a very interesting observation i hadn't thought about that but you're right it's they have they arrived with one script but where they're in an environment where there were multiple scripts that could be used and there were surprise and it's how they dealt with surprise on different scripts that led to the conflict which is kind of interesting as well with token who who's in a sense created the middle kingdom that it's filled full of piggies and and ralph's but the other characters aren't there jack is not there in tokens idealized communication of man and nature so there there is that interesting juxtaposition of lord of the flies and lord of the rings and one which um i think certainly um had more influence in terms of a scholarly and studied approach to personal interactions you know which you know you know i should not have been surprised you know roll forward 10 years and i was studying university and then subsequently psychiatry as a practicing uh practicing clinician so you know there was a richness there that was um possibly underwritten to a certain extent by reading those those books and you know another book i remember was animal farm around that time which seemed to have the same sense to it you know that you know this is this is the the integration of the psyche at an interpersonal level that could be afforded if you're if all those constraints don't you know aren't in play or that harmony that tolkien was restoring in his long story and was not was not you know attained yeah it's uh again you know looking at those books that we've discussed it really is a question of of harmony's juxtaposition with power and does power ultimately capture harmony for its own purposes and defines it in such a way in an orwellian way that you have it has the best of both worlds and everyone else has a script which is the worst of all worlds but it's their script and they believe in it so there there is i'm glad we're we're working on this metaphor of scripting because in a sense what we've been discussing uh in this conversation are the various that appear in the the literature here the diversity of them how in some ways it's the same script but in different contexts yes you know suddenly in lord of uh of the flies you have adolescent boys without supervision in a state of nature what kind of disintegration happens and what scripts arise out of catcher in the rye by j.d salinger an american novel a classic which is on your list again about an adolescent going through a particularly difficult psychological time being kicked out of his exclusive school and private school in pennsylvania and making his mission to save all the innocent children of the from going off the cliff through ignorance or their own innocence so what drew you to this particular book that was the book the last book i read so i've been in in my late teens if not early 20s then and um i i can't remember how i came across it but you know i i do remember reading it while traveling and i don't remember where i was traveling but becoming absolutely

engrossed so that was one of those books you just couldn't put down i don't know why with certain grossing it was um it had the same um the same kind of completeness that lord of the rings brings to the table in terms of defining a complete universe where you're all the rules are known nothing is resolved but all the contingencies are in play and indeed there was a little map in my version of the rings you could actually see where you were that was bounded um and in the same subtle way but in a different way um i think catching the rye had that that this was you know a little universe which was i i can't remember now because it's as i say more than half a century since many decades since i read it but it had that it was is this is a little island a little bounded island of a few days in a young man's life where the earth just got into that little island that little sort of epoch it was completely self-contained an episode and you didn't have to worry about the you know the you know the past or the future you just were in this particular bounded um critique description uh adventure within this within this epoch and that was very comforting but also very frightening because again there's a sense of disintegration so i probably misread it um what i what i remember from this was was and it probably is again a made memory due to my psychiatric training later in life but i remember it as basically uh what we'd now describe as an early s you know an early psychotic break a young man um experiencing the first psychotic break and disintegration again of the psyche the the disconnection of the the connectivity that ties our world models together in inside our head and all the um the delusional mood and the misperception or misperception of of what matters and what's going on in just daily life and daily transactions with you know with relatives and friends and and teachers and indeed i think one point ends up in a psychiatric asylum yeah yeah yeah um so i remember it acutely because you know because of course i within within a few years i was actually dealing with these people uh you know students at oxford university who were experiencing the early um symptoms of going through very similar episodes and weekends and you know so it was just a very poignant very personal i thought it was almost autobiographical but you know obviously it wasn't it read as if it was yeah yeah again had it had an honesty in its disintegration and its chaoticness with which you know you know as a student of the way things could go wrong um you know it was it was very engaging so it was obviously influential as well uh in terms of career path and in terms of your your own modeling of the world particularly adolescence in the world would be in a way modeled by holton caulfield and his uh inability uh to kind of accept uh a larger reality there's one quote i'd like to share with you from from the book uh which shows the quality of the writing and i i think uh the essence of the story she wasn't doing a thing i could see except

standing there leaning on a balcony railing holding the universe together that to me is the epitome of great art putting the mundane with the sublime the local with the universal the ordinary with the meta in one simple sentence leaning against the balcony somehow making that kind of a cosmic connection between two very disparate again you hear your counterfactuals yes put together in one sentence in order to make a profound point of the interconnection that goes on between counterfactuals and and perhaps that it was that style of writing and that beautiful juxtaposition of things which just could not normally go together within our sort of you know our normative appreciation of the world which made me think it was autobiographical i mean to have that ability not dissimilar to the sort of lateral thinking and certainly almost formally identical to um night's move thinking or um derailment schizophrenia where things your things jump about and there are collisions and juxtapositions and associations that are completely implausible in one's normal internal narrative or indeed spoke spoken the kind of the kind of thing that exactly is the um required to build new models to to be creative so you know i imagine it's kind of thing that van gogh had in a visual context absolutely yeah the the um the other thing about that particular example about the balcony and holding the universe together is that it has this sort of um grandiosity of uh you know which again is a hallmark of certain kinds of psychosis so manic depressive psychosis that you know the the the the true or false impression that your your your your existence your matters in terms of you know um in a uh you know in a manic sense omnipotent sense matters in terms of the existence of the but also um notice within that holding the universe together that your influence transcends or encompasses the entire universe a complete loss of boundary you know there is no boundary between you and then you get into notions for your notions of ego dissolution but of course given our previous conversation the dissolution of the markov blanket yes you're responsible for holding the universe together yes exactly the the blanket dissolved but therefore a fraction of a nanosecond it exists and you see something you haven't quite seen before it's interesting uh that line between and schizophrenia where imagination at some point along a continuum is no longer a healthy vehicle to resolve uncertainty it is a vehicle that's basically the steering wheel's coming off and the brakes don't work yes yeah i think it's a very nice way of putting it yep um so i mean just on logical arguments that that has to be right in the sense that we still have people with this kind of psychosis in our population so from an evolutionary point of view that speaks to the fact there is something very adaptive about having a schizotypal tendency you know there's something that promotes your ability to survive and to live in this culture in this world on an evolutionary time scale if you've got this

propensity for exploring counterfactuals but as you say you can go too far so there's something you know there has to be something slightly schizophrenic about all of us if we're going to make a difference or indeed survive in in this world um but the you know the when it goes too far which is that gets into the the which i think i think holden probably you know if i read that now and applied so you psychiatric criteria i think diagnosis but it all is all about coming back to um you know sort of the reduction of and inference it's all about inference i you know in a sense what are delusions and hallucinations if not simply falsely inferring something is there when it's not uh or what is an agnosia or a neglect syndrome where you you say you know this is not my arm or i can't recognize that that person's face so that's basically inferring something is not there when it is but in both instances it's it's it's a sort of a disorder of inference it's you're making the wrong sense of the sensory impressions um you know clearly we have to suspend our um commitment to the sensory evidence in order to imagine perhaps the most beautiful example this is dreaming so this is you know something very close to the heart of uh one of my um most senior and close colleagues in consciousness and sleep research alan hobson so if you want simple evidence the brain can construct its own realities and its own explanations for the world then you need to look no further than dreaming and that this is you know at the heart of consciousness and sentience uh but that sentence is a cons you know an active construction that the brain has generated where you use the sensory information during wakefulness at least to constrain and shape and do do the structural early a very delicate game and if it goes wrong you can easily see how you can fall into a state of psychosis um and indeed you can possibly even understand certain sleep disorders along those lines yeah that's interesting i haven't thought about that it makes sense because in a dream state the input from the world is is cut off so what is being filtered in is not from experience it's from the data banks of models in your head that then create their own without any kind of countervailing reality check with the evidence at hand yes it's so you can imagine that that's a nice example every night you put on a very very thick markov blanket i mean you can you know you're successful sleep is is a thick mark off blanket one that you can't throw off to cool down until you wake up it's there and dreams are the hallucination from inside a thick markov blanket alan hobson would love that i'll i'll tell him that okay so we're coming then to the uh human condition and halden caulfield and catcher in the rye where let me give you a couple of quotes to remind you some of the nice parts of this book the nature of truth it's partly true but it isn't all true people always think something's all true again one of those leaning on the balcony kind of statements the human condition among other things you find that you're not the first

one who was ever confused and frightened or even sickened by human behavior and that goes back to some of the earlier books on the list as well is that realization of the diversity of behavior and that there are things that are going on outside of york in places like catalonia that i was never aware of far from the maddening crowd thomas hardy that's a book i'm familiar with i went through a stage of loving thomas hardy read everything again i think a contemporary of d.h lawrence in writing about relationships uh between the classes and and particularly in far from the mounting crowd you have the wonderful uh bashiba and her estate a very accomplished self-reliant woman who suddenly decides that she will risk everything for a rogue named troy who really just cast his luck to the to the the fates and doesn't really take care of her but she's willing to risk everything for this even though there are people around him men around her who wish to court her but she will have nothing to do with them so we're seeing uh in a way a kind of self-destructive streak uh set in a class setting what was what was the the book far from the maddening crowd for you growing up it was um it was in part escapism of the sort that one enjoys with um with with tolkien and you know the other um you know so great fantasy like uh um or that inherits from the fact it was in in past times so you knew coming back that you knew there were going to be no nuclear explosions or walls so that you know the parameters were set but within those parameters there was you know intense tragedy you know i think that the you picking up on the self-destruction that that that was the tragedy and certainly in the context of all the men who who could have made it all right sure have the self-evident naughtiness of troy was in front of joy within the parameters of this little um in the southwest corner of a rural england um we sort of took on a magnitude which was quite horrifying you know you just couldn't believe it was getting worse and worse and how awful as troy was particularly from i do remember feeling very sorry for gabriel i do remember the opening the opening thing where he loses his shiva that was horrible as well because i i used to have animals i wanted to be a vet at one point um so this sort of um getting back to the yorkshire moors um aspiration was underneath a lot of this um but it was the it was for me again the perversity of um a failure to properly um develop a theory of mind or infer the intentions and the nature of another and by ascribing them inappropriately the attributes that you would have of sheba would have had to to troy she failed to infer what kind person he was and you know unfortunately had to witness the evidence towards the end of the story that she he wasn't the kind of person that she was so you know it's a failure of a shared narrative um in a in in a romantic and and interpersonal sort um which you know is irresistible uh you know as a story as long as it doesn't happen to me yeah you know in a way apart from the

maddening crowd you know the central heroine there you can see that the deficiency is she didn't have appropriate scripts to model the world of men and what the capabilities were and what the dangers were her motto was inadequate and she was unable to ask the right questions in order to update that model and so she just got deeper and deeper in holding on to the she knew even when it was clear that it was failing her she was unable to let it go yes yes yes i mean naivety um of the sword that inherits from not having access to those counterfactuals so had she read thomas hardy she'd been fine he judged the obscure i would recommend so it uh the book kind of ends with the world is largely made up of compromise and the the compromise i take that to mean in a larger sense maybe a little bit like leaning on the balcony uh kind of metaphor is that the compromise is with your ability to control the scripts of others and your uh perhaps inattentiveness in updating your own scripts in order to have a more meaningful effective way of communicating others what you intend what your motives are what you wish to see happen that was wonderfully important in particular that notion of updating your own scripts as part of that i would say existentially necessary way of updating your world models so that actually that word updating plays a really central role in formal mathematical descriptions of model optimization structure and it's called bayesian belief updating it is literally um updating your belief states on the basis of evidence so you know this is this is com if compromise is basically being able to accumulate and assimilate evidence in the service of updating your belief spacing belief updating in order to improve or optimize your then compromises is if you like an entirely and complete description of the way that we live yes and and another one is perhaps a little darker is that the compromise of the updating of your scripts is also dependent on the power of whatever mondo you're living under that somehow the updating isn't equal across the board and i think it's important to understand that culturally that updating process is viewed in very different ways and has been done historically where the power is the environment it is absolutely terribly one would hope is not so you won't find uh you won't find an orwellian utopia in in wessex though this is why it's so nice to meet thomas hardy this has been a delightful conversation carl i very much appreciate you coming on the show and sharing with us your your in-depth knowledge of science of neurology of mapping of the human brain we can go on for a very long time you've added a lot the books you've chosen are classic and i hope that the viewers will appreciate the insights that you've brought to those books and to the general conversation on how we use books to model our world and how that model influences who we are what we are what we see and what we want that was a lovely closure i greatly enjoyed talking to you we talked about all

my favorite things thank you thank you sir you have a great day and i will let you know when
the show is up thanks a lot thank you cheers